Sequences (Part 1) – Worksheet
MCR3U

SOLUTIONS

General formula for an Arithmetic Sequence:

$$t_n = a + (n-1)(d)$$

General formula for a Geometric Sequence:

$$t_n = a \cdot r^{n-1}$$

1) Find the next three terms of each arithmetic sequence.

a) 3, 7, 11, 15, 19, 23, 27 (+4)

b) -13, -11, -9, -7, -5, -3, -1 (+2)

c) 22, 20, 18, 16, 14, 12, 10 (-2)

d) -2, -5, -8, -11, -14, -17, -20 (-3)

2) Find the next three terms of each geometric sequence.

a) 4, 8, 16, 32, 64, 128 (×2)

b) 1, -6, 36, -216, 1296, -7776 (×-6)

c) 486, 162, 54, 18, 6, 2 (×$\frac{1}{3}$)

d) 3, 15, 75, 375, 1875, 9375 (×5)

3) Determine whether each sequence is an arithmetic sequence, a geometric sequence or neither. If it is an arithmetic or geometric sequence, determine a formula to represent the sequence.

a) 4, 7, 9, 12, ...
neither

b) 15, 13, 11, 9, ... (-2)
Arithmetic
$t_n = 15 + (n-1)(-2)$

c) 4, 12, 36, 108, ... (×3)
Geometric
$t_n = 4(3)^{n-1}$

d) 5, 10, 15, 20, ... (+5)
Arithmetic
$t_n = 5 + (n-1)(5)$

e) 7, 10, 13, 16, ... (+3)
Arithmetic
$t_n = 7 + (n-1)(3)$

f) 120, -60, 30, -15, .. (×$-\frac{1}{2}$)
Geometric
$t_n = 120\left(-\frac{1}{2}\right)^{n-1}$

g) -6, -5, -3, -1, ...
Neither

h) -13, -6, 1, 8, ... (+7)
Arithmetic
$t_n = -13 + (n-1)(7)$

i) 625, 125, 25, 5, ... (×$\frac{1}{5}$)
Geometric
$t_n = 625\left(\frac{1}{5}\right)^{n-1}$

4) Charlie deposited $115 in a savings account. Each week thereafter, he deposits $35 into the account.

a) Write a formula to represent this sequence.

$t_n = 115 + (n-1)(35)$

b) How much total money has Charlie deposited after 30 weeks?

$t_{30} = 115 + (30-1)(35)$

$= 115 + 1015$

$= \$1130$

5) A ball is dropped from a height of 500 meters. The table shows the height of each bounce.

BOUNCE #	HEIGHT (m)
1	400
2	320
3	256

$\times \frac{4}{5}$ OR 0.8

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a) Write an equation to represent the height of the ball after each bounce.

$t_n = 400(0.8)^{n-1}$

b) How high does the ball bounce on the 6th bounce?

$t_6 = 400(0.8)^{6-1}$

$= 131.072 \text{ m}$

Answers

1 a) 3, 7, 11, 15, 19, 23, 27,… **b)** -13, -11, -9, -7, -5, -3, -1,…
c) 22, 20, 18, 16, 14, 12, 10,… **d)** -2, -5, -8, -11, -14, -17, -20,…

2) a) 4, 8, 16, 32, 64, 128,… **b)** 1, -6, 36, -216, 1296, -7776,…
c) 486, 162, 54, 27, 9, 3,… **d)** 3, 15, 75, 375, 1875, 9375,…

3) a) 4, 7, 9, 12, … neither
b) 15, 13, 11, 9, … arithmetic $t_n = 15 +(n-1)(-2)$
c) 4, 12, 36, 108, … geometric $t_n = 4(3)^{n-1}$
d) 5, 10, 15, 20, … arithmetic $t_n = 5 +(n-1)(5)$
e) 7, 10, 13, 16, … arithmetic $t_n = 7 +(n-1)(3)$
f) 120, -60, 30, -15, ... geometric $t_n = 120(-1/2)^{n-1}$
g) -6, -5, -3, -1, … neither
h) -13, -6, 1, 8, … arithmetic $t_n = -13 +(n-1)(7)$
i) 625, 125, 25, 5, … geometric $t_n = 625(1/5)^{n-1}$

4) a) $t_n = 115 +(n-1)(35)$ **b)** $t_{30} = 1130$

5) a) $t_n = 400(0.8)^{n-1}$ **b)** $t_6=131.072$